



TEST & MEASUREMENT



Fig. 3: OTU-5000 optical test unit by Viavi Solutions

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long distances with a small amount of backscattered light using sensitive receivers and signal averaging.

Speed of the pulse can be calibrated as it passes down the fibre to create a display of the amount of backscattered light at any point in the fibre. Thus, an OTDR can measure time, calculate the pulse position in the fibre and correlate to what it sees in backscattered light with an actual location in the fibre.

Amount of power in the test pulse decreases in the cable plant under test, since it is attenuated as it passes along the fibre and suffers loss in connectors and splices. Thus, backscattered light reduces accordingly, producing a picture of the actual loss occurring in the fibre.

Slope of the fibre trace shows the attenuation coefficient (dB/km) of the fibre. Height of the peak indicates the amount of reflection at the connectors and splices. If peak has a flat top and tail on the far end, it indicates that the receiver is overloaded. Width of the peak signifies the distance resolution of the OTDR. An OTDR can also detect problems in the cable caused during installation. Avoid measuring loss in a section where the fibre looks non-linear at either end, especially near a reflective connector or splice.

Viavi Solutions' OTU-5000 optical test unit combines OTDR and optical-switch technology to provide continuous monitoring of multiple fibres anywhere in the network. A single unit monitoring 72 fibres of 100km and more occupies only 1RU. It offers Web browser access, switch scalability up to 1080 ports, fault location detection and email notifications. It has dual power feeds, low power consumption and LAN-based firmware download features.

Need for certification

Certification of fibre-optic cabling as per IEEE, TIA/EIA or ISO/IEC standards is necessary to ensure that the link will run the desired application. It includes two parts: basic test regimen and extended test regimen. Basic fibre optic cabling certification is performed with a power meter and light source or optical loss test set to ensure that the cable plant is within the loss budget before acceptance of the installation. And, extended fibre optic cabling certification and troubleshooting can be performed with an OTDR.

Insertion loss of an installed fibre-optic cable plant is the most important to measure. It is measured with a light source and power meter or optical loss test set, which is required by all international standards. Fibre-optic certification utilises an OTDR for loss testing to ensure the quality of individual components of the installed link. Fibre-optic cabling certification testing has become more important with an increase in network speeds, bandwidth demands, reduced distances and loss limitations.

Certification of fibre links requires the right testing equipment, and detailed knowledge of installation and application standards. A handheld fibre-optic certification tester, such as Fluke Networks' CertiFiber Pro, can quickly and easily certify multi-mode and single-mode networks. It measures fibre length and optical loss on two fibres at two wavelengths. It can also compute optical loss budget, compare the results to the selected industry standards and provide an instant pass or fail indication. This fibre certifier's test results can easily be saved and managed using such software as LinkWare (in case of CertiFiber Pro). **EFY**